

Engineering Statistics

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Chapter 10: Design of Experiments

Chapter Goals

- ▶ Terminology of experiments
- ▶ Randomized, comparative experiments
- ▶ Completely randomized designs
- ▶ The elements of the good experiment

Terminology of Experiments

- ▶ The individuals in an experiment are the **experimental units**. If they are human, we call them **subjects**.
- ▶ The explanatory variables in an experiment are often called **factors**.
- ▶ A **treatment** is a specific experimental condition applied to the subjects. With several factors, it is a combination of specific levels of each factor.
- ▶ Example: Compare blood pressure after six months between people on a diet/exercise program and people not on the program.

Benefits of Experimental Designs

- ▶ Observational studies are convenient but often subject to bias.
- ▶ Comparing observational groups does not establish that group differences cause different outcomes.
- ▶ Experiments provide more control:
 - ▶ Assign a treatment to subjects.
 - ▶ Hold environmental factors constant when they are not of direct interest.
- ▶ These features limit confounding and support arguments about causation.
- ▶ Experiments can study the combined effects of several factors.

Randomized, Comparative Experiments

- ▶ Gastric freezing was once recommended to treat peptic ulcers: patients swallowed a balloon through which refrigerated liquid was pumped to cool the stomach.
- ▶ Patients reported reduced pain, and the treatment was widely used.
- ▶ In the initial design, all patients received the treatment; pain reduction was recorded.
- ▶ **Questions:**
 - ▶ What would have happened if patients had been left alone?
 - ▶ Did patients feel better because of the care they received?

Confounding in the Gastric-Freezing Example

- Improvement could be confounded with natural recovery and psychological improvements from medical attention.

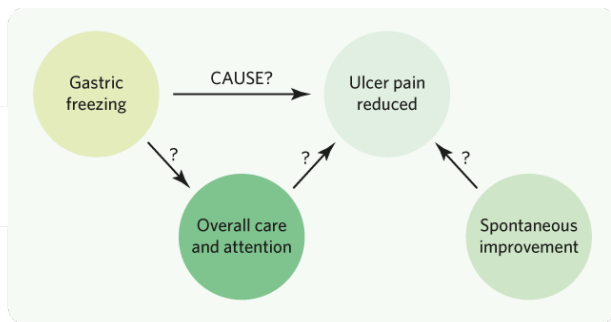
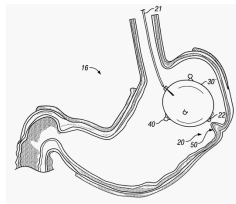
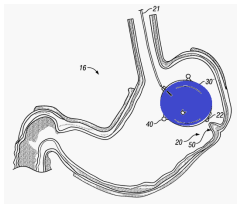


Figure 7.2

Baldi/Moore, *The Practice of Statistics in the Life Sciences*, 4e, © 2018 W. H. Freeman and Company

Randomized, Comparative Experiments: Design

1. Select a group of patients, say 16 people.
2. Prepare two types of gastric-freezing balloons: one with refrigerated liquid and one without, eight of each.



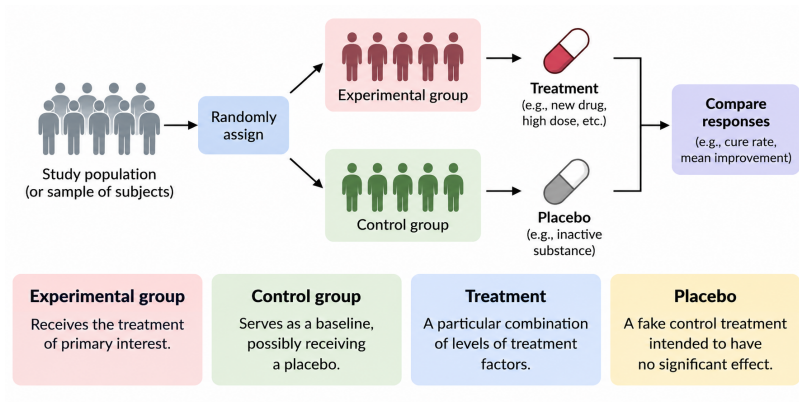
Randomized, Comparative Experiments: Results

3. Randomly assign eight patients to each treatment using a coin.
 4. Measure the reduction in pain.
 5. Use a statistical method to compare the groups.
- ▶ In a randomized comparative experiment, 28 of 82 gastric-freezing patients improved.
 - ▶ In the control group, 30 of 78 patients improved.
 - ▶ Because the improvement rates were similar, gastric freezing was abandoned.

Terminology Again!

- ▶ **Experimental group:** receives the treatment of primary interest.
- ▶ **Control group:** serves as a baseline, possibly receiving a placebo.
- ▶ **Treatment:** a particular combination of levels of treatment factors.
- ▶ **Placebo:** a fake control treatment intended to have no significant effect.
- ▶ **Question:** Does the way we assign treatments to individuals matter?

Key Concepts in Experimental Design



Terminology: Gastric-Freezing Example

- ▶ **Treatments:** balloon with refrigerated liquid; balloon without refrigerated liquid.
- ▶ **Placebo:** balloon without refrigerated liquid.
- ▶ **Experimental group:** patients receiving refrigerated liquid.
- ▶ **Control group:** patients receiving the fake treatment.

Well-Designed Experiment

- ▶ **Key point:** Randomly assign subjects to experimental conditions, then observe the response outcomes.
- ▶ Important terminology:
 - ▶ Response and explanatory variables
 - ▶ Treatment and blocking factors
 - ▶ Control group, treatment, and placebo
 - ▶ Observational and experimental units

Example #1: Exercise and Diet

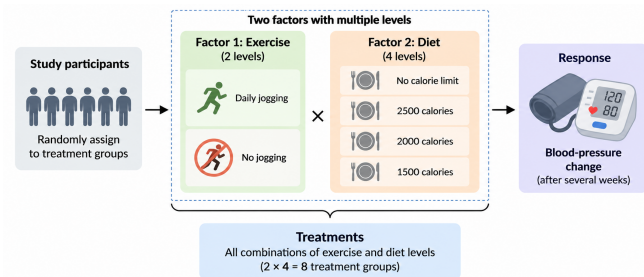
- ▶ Conduct an experiment measuring blood-pressure change after several weeks of an exercise and diet regimen.
- ▶ **Response:** blood-pressure change.
- ▶ **Factor 1 — Exercise:** daily jogging; no jogging.
- ▶ **Factor 2 — Diet:** no calorie limit; limits of 2500, 2000, or 1500 calories.
- ▶ There is one response variable and two explanatory variables. Each factor has different **levels**.

Example #2: Exercise, Diet, and Gender

- ▶ Measure blood-pressure change using the same exercise and diet regimen, while also considering gender.
- ▶ **Exercise:** daily jogging or no jogging.
- ▶ **Diet:** no calorie limit or limits of 2500, 2000, or 1500 calories.
- ▶ **Gender:** male or female.
- ▶ **Treatment factor:** researcher assigns its levels to subjects.
- ▶ **Blocking factor:** levels are determined by subject characteristics, such as demographics.

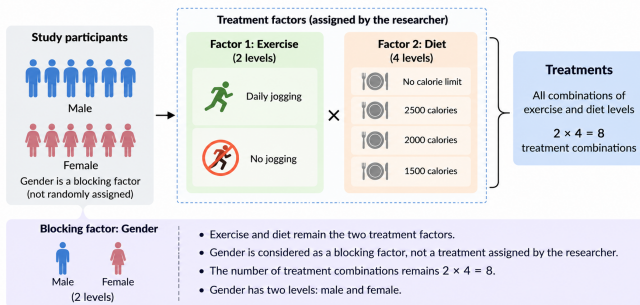
Treatments in Example #1

- ▶ A treatment is a combination of one level of each treatment factor.
- ▶ Exercise has two levels: daily jogging or no jogging.
- ▶ Diet has four levels: no limit, 2500, 2000, or 1500 calories.
- ▶ **Number of treatments:** $2 \times 4 = 8$.
- ▶ In general, multiply the numbers of levels of all treatment factors.



Treatments in Example #2

- ▶ Exercise and diet remain the two treatment factors.
- ▶ Gender is considered as a blocking factor, not a treatment assigned by the researcher.
- ▶ The number of treatment combinations remains $2 \times 4 = 8$.
- ▶ Gender has two levels: male and female.



Treatment Combinations

- ▶ Both examples have eight treatments.
- ▶ No jogging with: no calorie limit, 2500, 2000, or 1500 calories.
- ▶ Daily jogging with: no calorie limit, 2500, 2000, or 1500 calories.
- ▶ After determining the treatments, randomly assign people to them.

No jogging and No calories limits	No Jogging and limit 2500	No jogging and limit 2000	No jogging and limit 1500
Daily jogging and No calories limit	Daily jogging and limit 2500	Daily jogging and limit 2000	Daily jogging and limit 1500

Control Group

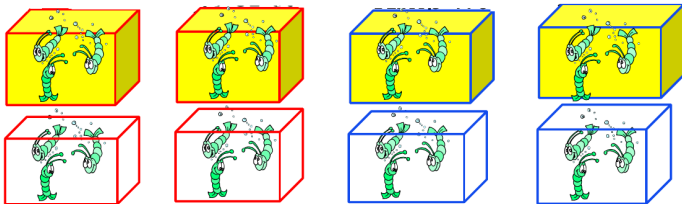
- ▶ The control group receives the combination of baseline levels of the treatment factors.
- ▶ In the exercise-and-diet example, the control group receives:
 - ▶ No jogging.
 - ▶ No calorie limit.

Experimental Units versus Observational Units

- ▶ **Experimental units (EUs):** individuals to which treatments are applied.
- ▶ **Observational units (OUs):** individuals from which responses are measured.
- ▶ In many experiments, EUs and OUs are the same, but this is not always true.

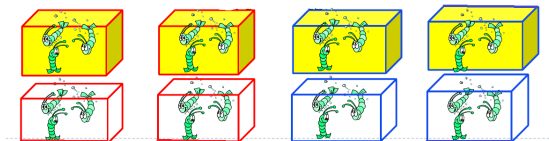
Example #3: Shrimp Growth

- ▶ An experiment measures the growth of shrimp.
- ▶ Equal numbers of shrimp are placed in eight tanks with assigned temperature and salinity levels.
- ▶ After several weeks, the weight change of each shrimp is recorded.
- ▶ Red/blue denote high/low temperature; yellow/white denote salty/not salty.



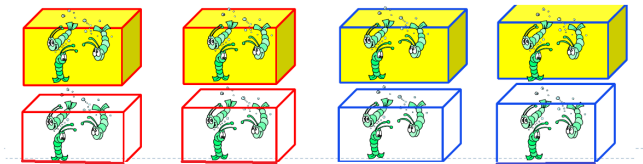
Example #3: Identify the Design Elements

- ▶ **Experimental units:** tanks; **observational units:** shrimp.
- ▶ **Response:** growth (weight change).
- ▶ **Factors:** salinity and temperature.
- ▶ **Blocking factors:** none.
- ▶ **Treatment factors:** two.
- ▶ **Possible treatments:** four.
- ▶ **Shrimp assigned per tank:** six.



Example #3: Experimental and Observational Units

- ▶ The experimental units are the tanks.
- ▶ The observational units are the shrimp.
- ▶ **Number of observational units: 24.**
- ▶ **Number of experimental units: 8.**



Completely Randomized Designs

- ▶ All factors are treatment factors, and subjects are randomly assigned to treatments.
- ▶ Each subject receives one treatment.
- ▶ In the exercise-and-diet example, there are eight treatments.
- ▶ With 24 subjects, randomly assign three people to each treatment.

The Elements of a Good Experiment

- ▶ **Control group:** provides a baseline, possibly receiving a placebo.
- ▶ **Randomization:** randomly assign subjects to treatments to reduce bias and potential confounding.
- ▶ **Replication:** assign more than one subject to each treatment.

The Elements of a Good Experiment: Blinding

- ▶ **Blinded experiment:** subjects do not know their treatment assignments.
- ▶ **Double-blinded experiment:** neither subjects nor researchers know the assignments.
- ▶ A third party may make assignments and conceal the treatment labels.
- ▶ Blinding and double-blinding reduce bias.

Statistical Analysis of Experimental Data

- ▶ **Goal:** Determine whether treatments have an effect on the response variable.
- ▶ **Summarize the data.** Compute sample means, proportions, and measures of variability for each treatment group.
- ▶ **Visualize treatment differences.** Use boxplots, histograms, or other graphs to compare groups.
- ▶ **Quantify uncertainty.** Use confidence intervals to estimate treatment effects.
- ▶ **Test for treatment effects.** Use hypothesis tests to determine whether observed differences could be explained by random variation.

Choosing a Statistical Method

- ▶ **Two treatment groups, continuous response:** Use a two-sample t -test to compare population means.
- ▶ **Two treatment groups, binary response:** Use a two-proportion Z -test to compare success rates.
- ▶ **Paired or matched observations:** Use a paired t -test to analyze within-pair differences.
- ▶ **More than two treatments or multiple factors:** More advanced methods are needed to analyze treatment effects and interactions.

Next: ANOVA and more advanced experimental analysis methods.

Online Experiments

- ▶ **Online experiments** are randomized experiments conducted on digital platforms, such as websites and mobile apps.
- ▶ Users are randomly assigned to different versions of a product, feature, or service.
- ▶ The goal is to measure the **causal effect** of a change on user behavior.
- ▶ Common response variables include:
 - ▶ Click-through rate
 - ▶ Conversion rate
 - ▶ Revenue per user
 - ▶ User engagement

A/B Testing

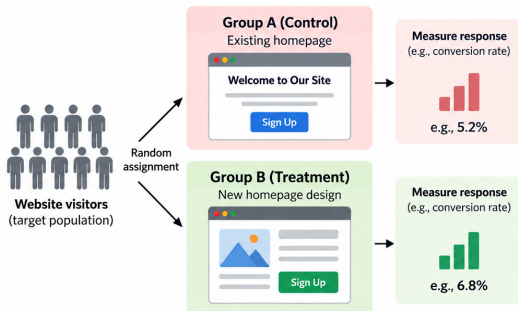
- ▶ **A/B testing** is a simple online randomized comparative experiment.
- ▶ Users are randomly assigned to two groups:
 - ▶ **Group A (Control)**: existing version.
 - ▶ **Group B (Treatment)**: modified version.
- ▶ Example: Does changing a website's checkout button increase the purchase conversion rate?
- ▶ Compare the response variables between the two groups to estimate the treatment effect.

Random Assignment → A/B Groups → Compare Outcomes

Statistical Analysis of A/B Tests

- ▶ **Define the objective before the experiment.** Choose the response variable and the effect of interest.
- ▶ **Randomize users and collect data.** Ensure comparable groups and sufficient sample sizes.
- ▶ **Estimate and test the treatment effect.** Use confidence intervals and hypothesis tests to compare the two groups.
- ▶ **Distinguish statistical and practical significance.** A statistically significant improvement may be too small to matter in practice.
- ▶ **Avoid common pitfalls.** Repeatedly checking results and stopping when significant can inflate false-positive rates.

A/B Testing: Learning with AI



How AI Can Help



Learn concepts

- Explain A/B testing with examples
- Summarize key ideas (randomization, control group, treatment effect, etc.)



Generate code

- Provide R/Python code to analyze A/B test data
- Explain the code step by step



Interpret results

- Help interpret output (e.g., p-value, confidence interval)
- Explain practical significance



Explore further

- Suggest best practices
- Discuss common pitfalls (e.g., peeking, multiple testing)
- Recommend additional resources

Key Takeaways

- Online experiments apply the same principles as classical experimental design.
- Randomization helps ensure comparable groups and supports causal conclusions.
- A/B testing is a widely used and practical example.
- AI tools can be helpful for learning concepts, generating code, and interpreting results, but you should always verify the results and ensure they are appropriate for the problem.

*Use AI as a learning assistant,
not a substitute for understanding.*

Chapter Summary

- ▶ **Use experiments to investigate causal relationships.**
Assign treatments to experimental units and measure their effects on response variables.
- ▶ **Apply the four main principles of experimental design.**
 - ▶ **Control:** Use baseline groups and control experimental conditions.
 - ▶ **Randomization:** Randomly assign treatments to reduce bias.
 - ▶ **Replication:** Use multiple experimental units to assess variability.
 - ▶ **Blocking:** Group similar units to reduce unwanted variability.
- ▶ **Use statistical methods to evaluate treatment effects.**
Summarize and visualize data, then use confidence intervals and hypothesis tests to assess treatment differences.
- ▶ **Apply DOE principles to online experiments.** A/B testing uses random assignment and comparison groups to evaluate changes in digital products and services.